

Comprehensive Analysis of Coral Skeletal Geochemistry: Stable Isotope Dynamics and Trace Element Proxies in Paleoenvironmental Reconstruction

The scientific investigation of scleractinian coral skeletons has emerged as a cornerstone of paleoclimatology, providing high-resolution, multi-century records of the tropical oceans' physical and chemical state. Corals, particularly massive species such as those within the genus *Porites*, serve as unique environmental archives because they grow continuously, produce annual density bands similar to tree rings, and incorporate a suite of isotopic and elemental tracers from the surrounding seawater.¹ The fidelity of these records is predicated on a sophisticated understanding of the biomineralization process, a biologically mediated mechanism that precipitates aragonite from a semi-isolated calcifying fluid.¹ This report examines the intricate relationships between stable isotopes—specifically $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ —and trace element ratios such as Sr/Ca and Ba/Ca in relation to environmental variables including sea surface temperature (SST), sea surface salinity (SSS), solar radiation, and the anthropogenic carbon cycle.

Biomineralization Mechanisms and the Physicochemical Environment

Scleractinian corals precipitate their skeletons of aragonite, an orthorhombic polymorph of calcium carbonate (CaCO_3), through the activity of specialized cells known as the calicoblastic ectoderm.¹ This process occurs within an extracellular calcifying fluid (ECF) situated between the base of the coral polyp and the existing skeletal surface.¹ The coral polyp isolates a volume of seawater and actively manipulates its chemistry to create conditions highly supersaturated with respect to aragonite, facilitating mineral nucleation and growth.⁵

The Role of Endosymbiotic Zooxanthellae

The physiology of reef-building (hermatypic) corals is inextricably linked to their symbiotic relationship with dinoflagellate algae, or zooxanthellae, residing within the coral's endoderm.¹ These symbionts perform photosynthesis, which significantly augments the rate of skeletal calcification—a phenomenon often referred to as light-enhanced calcification.¹ The presence of zooxanthellae influences the skeletal geochemistry through several pathways, including the translocation of photosynthate-derived carbon to the polyp and the modification of the

internal dissolved inorganic carbon (DIC) pool's isotopic composition.¹

Microstructure and Geochemical Heterogeneity

Coral skeletal microstructure reveals a complex matrix of aragonitic needles and organic components.¹ High-resolution micro-analytical measurements have demonstrated that different skeletal components, such as the centers of calcification (COC) and the surrounding fibrous bundles, possess distinct geochemical signatures.⁶ COCs are typically characterized by higher trace element concentrations and more depleted isotopic compositions than the fibrous skeleton.¹⁰ These differences necessitate the existence of multiple calcification mechanisms controlled by specialized domains of the calicoblastic cell layer.¹⁰

Skeletal Component	Growth Characteristic	Geochemical Signature	Biom mineralization Mechanism
Centers of Calcification (COC)	Rapid, granular nucleation	Depleted $\delta^{13}C$ and $\delta^{18}O$, High trace elements	Biologically controlled amorphous precursor phase
Fibrous Skeleton	Slower, needle-like growth	Enriched $\delta^{13}C$ and $\delta^{18}O$, Lower trace elements	Physicochemical precipitation from ECF
Annual Density Bands	Seasonal variation	Fluctuating Isotope/Trace Element ratios	Environmental modulation of growth rate

The identification of these microstructural variations is critical for paleoclimate reconstruction, as bulk skeletal analyses integrate these heterogeneous signals. Standard practice involves sampling along the primary growth axis of massive corals to minimize the influence of secondary calcification and lateral growth variations.¹

Geochemistry of Stable Carbon Isotopes ($\delta^{13}C$) in Coral Skeletons

The stable carbon isotopic composition of coral skeletons ($\delta^{13}C$) provides a multifaceted record of biological activity and environmental change. Unlike oxygen isotopes, which are

primarily influenced by temperature and seawater composition, $\delta^{13}\text{C}$ is heavily modulated by coral metabolism, including the balance between photosynthesis and respiration (the P/R ratio) and the rate of calcification.⁹

Metabolic Fractionation and Light Sensitivity

Photosynthesis by endosymbiotic zooxanthellae is the dominant control on skeletal $\delta^{13}\text{C}$ levels. Photosynthetic organisms preferentially consume the lighter ^{12}C isotope, which enriches the internal DIC pool in the heavier ^{13}C isotope.⁹ As the rate of photosynthesis increases—typically in response to higher solar radiation—the skeletal $\delta^{13}\text{C}$ becomes more positive.⁹ This relationship makes coral $\delta^{13}\text{C}$ a potential proxy for cloud cover, water turbidity, and seasonal changes in solar insolation.¹¹

Conversely, respiration by the coral host and its symbionts releases isotopically light CO_2 back into the calcifying fluid, which can deplete the $\delta^{13}\text{C}$ of the skeleton.⁴ Heterotrophic feeding also introduces ^{13}C -depleted carbon into the coral's metabolic system, further influencing the skeletal signal.⁹ Depth-dependent studies have quantified this sensitivity, showing that mean coral $\delta^{13}\text{C}$ decreases by approximately -1.4‰ for every 5-meter increase in water depth, directly mirroring the attenuation of light and the subsequent reduction in photosynthetic rate.⁹

Kinetic Isotope Effects and Growth Rates

The "kinetic model" of coral calcification, popularized by McConnaughey, suggests that the rate of calcification controls $\delta^{13}\text{C}$ through fractionation at the rate-limiting step of CO_2 hydration or hydroxylation.⁴ In this framework, rapid skeletal growth prevents the DIC pool from reaching isotopic equilibrium with seawater, resulting in a depletion of both $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$.⁴

Extensive analysis of *Porites* corals from sites such as Fiji, Tonga, and American Samoa has revealed a consistent inverse relationship between skeletal $\delta^{13}\text{C}$ and linear extension (growth) rates over several centuries.⁹ Decadal-scale increases of 0.5–1.0‰ in $\delta^{13}\text{C}$ are frequently linked to significant reductions in growth rate, suggesting that kinetic fractionation modulates the $\delta^{13}\text{C}$ value around a mean determined by photosynthesis.⁹

The Anthropogenic Suess Effect and Carbon Cycle Trends

The most significant contemporary trend in coral $\delta^{13}\text{C}$ is the breakdown of the historical relationship between growth rate and carbon isotopes, driven by the atmospheric ^{13}C Suess effect.⁹ Since the mid-20th century, the global combustion of fossil fuels and extensive deforestation have released massive quantities of ^{13}C -depleted CO_2 into the atmosphere.⁹ This anthropogenic carbon is transferred to the surface ocean via gas exchange, lowering the $\delta^{13}\text{C}$ of the marine DIC pool.⁹

Coral skeletal records from globally distributed sites show a pronounced downward trend in $\delta^{13}\text{C}$ beginning between 1950 and 1960.⁹ For example, composite time series from Fiji demonstrate a long-term decrease of approximately 1.5‰ over 221 years, far exceeding what can be explained by natural changes in coral growth or metabolism.⁹ This shift often results in a decoupling where $\delta^{13}\text{C}$ continues to decline even when skeletal extension rates remain constant.⁹

Environmental Driver	Impact on $\delta^{13}\text{C}$	Primary Mechanism	Magnitude/Scale
Solar Radiation	Increase	Enrichment via photosynthesis (^{12}C consumption)	Seasonal/Interannual
Water Depth	Decrease	Depletion via light attenuation	−1.4‰ per 5m
Calcification Rate	Decrease (Inverse)	Kinetic fractionation during fast growth	0.5–1.0‰ deviations
Fossil Fuel Emissions	Decrease	Oceanic uptake of anthropogenic ^{12}C	> 1.5‰ since 1950
Bleaching Stress	Decrease	Loss of symbionts/photosynthesis	Abrupt, short-term

Stable Oxygen Isotopes ($\delta^{18}\text{O}$) as Temperature and Hydrology Proxies

The oxygen isotopic composition of coral skeletons ($\delta^{18}\text{O}$) is the primary tool for reconstructing past tropical climate variability. Its utility arises from its sensitivity to two fundamental environmental parameters: sea surface temperature (SST) and the oxygen isotopic composition of seawater ($\delta^{18}\text{O}_{sw}$), the latter of which tracks the local evaporation-precipitation balance.²

Thermodynamic Fractionation and Temperature Sensitivity

The fractionation of oxygen isotopes between seawater and the aragonite skeleton follows a predictable inverse relationship with temperature.³ As seawater temperatures rise, the incorporation of the heavier ^{18}O isotope into the skeleton decreases. Although coral skeletons are typically depleted in ^{18}O by several permil relative to inorganic aragonite precipitated at equilibrium, the slope of this temperature dependency is remarkably consistent with non-biological systems.³

Empirical studies have established a standard temperature response of approximately -0.18 ‰ to -0.22 ‰ per 1°C increase in SST.³ This high degree of sensitivity allows corals to record seasonal temperature cycles and interannual climate phenomena like the El Niño-Southern Oscillation (ENSO) with monthly resolution.⁴ However, inter-colony offsets of up to 0.3 ‰ have been observed between corals from the same location, potentially arising from micro-environmental differences or species-specific "vital effects".³

Hydrology, Salinity, and the $\delta^{18}\text{O}_{sw}$ Signal

The second major component of the skeletal $\delta^{18}\text{O}$ signal is the $\delta^{18}\text{O}$ of the ambient seawater.¹⁴ Seawater $\delta^{18}\text{O}$ values are influenced by the global water cycle, where evaporation enriches the surface ocean in ^{18}O and precipitation (which is isotopically light) depletes it.¹⁸ In the tropical oceans, $\delta^{18}\text{O}_{sw}$ exhibits a robust linear relationship with sea surface salinity (SSS).¹⁴

Reconstructing salinity from coral $\delta^{18}\text{O}$ requires the isolation of the $\delta^{18}\text{O}_{sw}$ signal by

removing the temperature component.² This is achieved through paired measurements of $\delta^{18}\text{O}$ and a "pure" temperature proxy, most commonly the strontium-to-calcium (Sr/Ca) ratio.²

Case Study: Indonesian Throughflow and Timor Sea Records

The application of paired $\delta^{18}\text{O}$ and Sr/Ca measurements is particularly effective in complex oceanographic regions like the Indonesian Throughflow (ITF).¹⁴ In these passages, the seasonal cycles of temperature and salinity are often out of phase.¹⁴ For instance, at Timor Island, the warmest and wettest months are both associated with depletion in $\delta^{18}\text{O}$, but they occur at different times.¹⁴

Paired analyses have revealed that while Sr/Ca faithfully records the SST, the derived $\delta^{18}\text{O}_{sw}$ tracks the migration of low-salinity water masses and the influence of climate modes like the Indian Ocean Dipole (IOD).¹⁴ These reconstructions have demonstrated that the ITF's heat and freshwater transport are influenced by interannual variability, which is not always captured by low-resolution gridded satellite products.¹⁴

Trace and Minor Element Proxies: Sr/Ca, Ba/Ca, and Beyond

In addition to stable isotopes, the elemental chemistry of the coral skeleton provides critical insights into ocean properties. Divalent cations such as Sr^{2+} , Ba^{2+} , and Mg^{2+} are incorporated into the aragonite lattice through substitution for Ca^{2+} , a process governed by thermodynamic partitioning and physiological regulation.²¹

Strontium-to-Calcium (Sr/Ca) as an SST Benchmark

The Sr/Ca ratio is widely considered the most reliable "direct" measure of water temperature in corals.³ The incorporation of strontium is inversely correlated with seawater temperature, following the thermodynamic properties of the mineral aragonite.³ Analytical uncertainties in Sr/Ca measurements are generally equivalent to less than 0.5°C, making it comparable in precision to oxygen isotope analysis.³

The robust nature of Sr/Ca as a thermometer stems from the relatively constant Sr/Ca ratio of open-ocean seawater over the timescales of most coral reconstructions.³ However, this assumption may be compromised in near-shore environments or areas with significant

terrestrial input, such as the Florida Keys, where seawater Sr/Ca may vary.²⁵ Furthermore, extreme biological stress, such as mass bleaching, can induce Sr/Ca anomalies that deviate from the expected SST relationship, likely due to a reduction in the activity of the Ca-ATPase pump which regulates ion transport.¹

Barium-to-Calcium (Ba/Ca): Runoff, Upwelling, and Nutrient Cycling

The Ba/Ca ratio in coral skeletons serves as a powerful proxy for tracking the delivery of barium-rich waters to the reef.²⁰ Barium concentrations in the ocean exhibit a "nutrient-like" profile, with enrichment in deep waters and freshwater runoff compared to surface seawater.²⁰

1. **Riverine Discharge:** In coastal regions, coral Ba/Ca records track monsoonal flood plumes and sediment delivery.²⁰ Barium desorbs from suspended sediments in low-salinity estuarine environments, increasing the local Ba concentration.²⁰ Multi-decadal records from the Great Barrier Reef (GBR) have used Ba/Ca to document historical changes in land-use and river flux.²⁰
2. **Upwelling Dynamics:** At reefs located near the continental shelf edge, Ba/Ca oscillations reflect the seasonal upwelling of cold, nutrient-rich thermocline waters.²⁰ Analysis of corals from the Galapagos and the northern GBR has identified sustained Ba/Ca peaks during periods of intense upwelling, providing a record of ocean circulation and productivity.²⁰

Recent studies utilizing stable barium isotopes ($\delta^{138/134}Ba$) have revealed that Ba^{2+} ions are transported to the calcification site primarily through passive mechanisms, such as seawater leakage.³⁰ This differs from the active transport of Ca^{2+} and Sr^{2+} , explaining why Ba/Ca records can be influenced by the coral's metabolic elevation of calcium concentrations in the ECF.³⁰

Magnesium-to-Calcium (Mg/Ca) and Diagenetic Considerations

Although Mg/Ca has been proposed as a temperature proxy, its application is fraught with complications.³¹ Unlike Sr/Ca , magnesium incorporation appears to be heavily influenced by the secretion of the skeletal organic matrix and growth rate.³² X-ray Absorption Fine Structure (XAFS) data has indicated that Mg may not be strictly substituted into the aragonite lattice but rather hosted in a disordered material or associated with organic macromolecules.³³ This

lack of structural consistency makes Mg/Ca a questionable proxy for temperature in fossil corals, although it remains useful as a marker for physiological processes and the identification of secondary calcite.²¹

Elemental Ratio	Environmental Proxy	Calibration Slope (typical)	Reliability/Issues
Sr/Ca	SST	-0.04 to -0.01 mmol/mol/°C	High; Affected by bleaching
Ba/Ca	Runoff / Upwelling	Site-specific	High; Site selection is critical
Mg/Ca	Calcification Rate	Variable	Low for SST; High "vital effects"
U/Ca	pH_{cf} / SST	Variable	Moderate; Tracks internal pH
Li/Mg	SST	Variable	Emerging; Potential high-precision

Competing Models of Coral Biomineralization: Kinetic vs. Metabolic

A central debate in coral geochemistry concerns the mechanisms that drive "vital effects"—the systematic offsets of biogenic minerals from thermodynamic equilibrium. Two primary conceptual frameworks dominate the literature: the McConnaughey kinetic model and the Adkins/metabolic model.⁶

The McConnaughey Kinetic Framework

The kinetic model posits that the simultaneous depletion of $\delta^{13}C$ and $\delta^{18}O$ in coral skeletons is caused by the fractionation during the CO_2 hydration/hydroxylation reaction.⁴ In this view, the rate of calcification is the primary driver of isotopic variability.⁴ This model explains the high degree of covariance often observed between carbon and oxygen isotopes in corals grown under constant environmental conditions.⁴

The Adkins Metabolic and pH-Regulation Model

The Adkins model, developed initially from studies of deep-sea corals (*Desmophyllum dianthus*), focuses on the coral's active regulation of the ECF's pH.⁶ As the coral pumps protons out of the ECF, it raises the pH to facilitate aragonite precipitation.³⁶ This pH elevation shifts the speciation of the DIC pool towards carbonate ions (CO_3^{2-}), which are isotopically lighter in oxygen.⁴

The Adkins model explains the "kinked" relationship between $\delta^{13}C$ and $\delta^{18}O$ observed in many species, where carbon and oxygen isotopes decouple at high pH.⁶ It also incorporates the role of metabolic CO_2 diffusion into the ECF, which contributes ^{13}C -depleted carbon to the mineral.⁴ This model is supported by microsensor measurements and boron isotope analyses that confirm corals internally up-regulate pH by as much as 1 unit relative to seawater.³⁶

Rayleigh Fractionation and Abiogenic Comparisons

A third perspective, introduced by Cohen and Gaetani, utilizes insights from abiogenic aragonite precipitation experiments.⁵ They argue that many "vital effects" are actually predictable physicochemical consequences of precipitating crystals from a finite reservoir.⁵ As aragonite forms in the semi-isolated ECF, the removal of ions like Ca and Sr follows a Rayleigh distillation process, which continuously alters the composition of the remaining fluid.⁵ This model emphasizes that crystal growth rate and the degree of seawater exchange (leakage) are more important controls on skeletal chemistry than previously realized.⁵

Paleoclimate Applications: ENSO, Volcanism, and Ocean Acidification

The integration of these various proxies allows for the reconstruction of past climate dynamics with unprecedented detail, extending beyond the limits of instrumental records.²

El Niño-Southern Oscillation (ENSO) Variability

Corals are the "natural choice" for ENSO reconstruction due to their location within the centers of ENSO action.³⁸ Large-scale efforts like the CoralHydro2k project have compiled hundreds of records to identify robust patterns of ENSO over the last millennium.² These reconstructions have revealed that ENSO teleconnections are not stationary and have shifted in response to changes in mean climate state, such as the transition into the Little Ice Age.⁴¹

Recent data assimilation studies, which combine coral proxies with tree-ring data, have refined the understanding of ENSO's response to external forcing.³⁸ While tree rings often show an El Niño-like response following major volcanic eruptions, coral records suggest a more complex, less uniform response, highlighting the importance of using multiple archives to characterize climate sensitivity.³⁸

Boron Isotopes ($\delta^{11}B$) and the Ocean Acidification Proxy

The boron isotopic composition of the coral skeleton ($\delta^{11}B$) is a direct proxy for seawater pH and the coral's internal buffering capacity.¹² Boron exists in seawater as boric acid and borate ions, with the relative abundance and isotopic composition of these species being pH -dependent.¹² Corals primarily incorporate the borate ion into their skeletons.³⁶

Measurements of $\delta^{11}B$ have shown that corals maintain their calcifying fluid pH (pH_{cf}) at levels higher than ambient seawater, a mechanism that provides some resilience to ocean acidification.³⁶ However, significant bleaching events or extreme temperature stress can impair this regulation, leading to a drop in pH_{cf} and a corresponding shift in the $\delta^{11}B$ record.³⁶ This robustness of $\delta^{11}B$ to short-term physiological changes (aside from bleaching) makes it an excellent proxy for reconstructing the long-term decrease in ocean pH since the industrial revolution.³⁶

Emerging Frontiers: Clumped Isotopes and Advanced Databases

The frontier of coral geochemistry involves the development of absolute temperature proxies and the curation of massive global datasets.

Carbonate Clumped Isotope Thermometry (Δ_{47})

Clumped isotopes measure the thermodynamic preference of ^{13}C and ^{18}O to form bonds with each other in the carbonate lattice—a preference that is solely dependent on temperature and independent of the fluid's isotopic composition.⁴⁴ While coral Δ_{47} values exhibit kinetic disequilibrium, "dual clumped isotope" thermometry (simultaneous Δ_{47} and Δ_{48} measurements) allows for the identification and correction of these offsets.⁴⁴ This technique has the potential to reconstruct both the SST and the absolute $\delta^{18}O_{sw}$ of past oceans,

providing a more direct measure of the global ice volume and hydrological cycle.⁴⁴

Data Curation and Integration

The utility of coral records in global climate modeling is limited by the accessibility and standardization of the data.¹⁵ Projects like CoralHydro2k and the Great Barrier Reef Skeletal Records Database (GBRCD) have transferred hundreds of peer-reviewed records into machine-readable formats with standardized metadata.² These databases track large-scale temperature and hydrological variability across the Common Era, facilitating comparisons with isotope-enabled climate model simulations and paleodata-assimilation projects.²

Conclusion: The Resilience and Vulnerability of the Coral Archive

Coral skeletal geochemistry provides a sophisticated, high-resolution record of the Earth's climate system, capturing the nuances of ocean warming, hydrological shifts, and the anthropogenic alteration of the carbon cycle. The $\delta^{13}\text{C}$ record offers a unique window into the P/R balance of reef ecosystems and the global Suess effect, while $\delta^{18}\text{O}$ and Sr/Ca provide the backbone for SST reconstruction. The use of secondary proxies like Ba/Ca and $\delta^{11}\text{B}$ allows for the disentanglement of complex environmental signals, including runoff, upwelling, and ocean acidification.

However, the continued reliability of these archives is challenged by the rapid decline of coral reefs due to climate change.⁴⁶ Mass bleaching events and ocean acidification not only threaten the survival of these organisms but also introduce physiological noise into the geochemical records.²⁶ As the world passes climate tipping points, the preservation and analysis of both modern and fossil coral skeletons remain imperative for understanding the full range of natural climate variability and the magnitude of the human impact on the oceans.⁴⁶ Through the integration of advanced biomineralization models, absolute temperature proxies like clumped isotopes, and robust global databases, coral geochemistry will continue to serve as a vital tool for navigating the challenges of the 21st century and beyond.

Works cited

1. Geochemistry of corals: Proxies of past ocean chemistry, ocean circulation, and climate - PMC, accessed February 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC33753/>
2. The CoralHydro2k database: a global, actively curated compilation of coral $\delta^{18}\text{O}$ and Sr / Ca proxy records of tropical ocean hydrology and temperature for the Common Era - ESSD, accessed February 23, 2026, <https://essd.copernicus.org/articles/15/2081/2023/>

3. Annually-banded corals as climate proxies., accessed February 23, 2026, <https://www.ncei.noaa.gov/pub/data/paleo/coral/corals.pdf>
4. MIT Open Access Articles Baseline shifts in coral ... - DSpace@MIT, accessed February 23, 2026, https://dspace.mit.edu/bitstream/handle/1721.1/104431/338_2012_Article_1004.pdf?sequence=1&isAllowed=y
5. Ion partitioning and the geochemistry of coral skeletons: solving the ..., accessed February 23, 2026, <https://www.whoi.edu/files/server.do?id=75643&pt=2&p=36767>
6. Carbonic anhydrase, coral calcification and a new model ... - Caltech, accessed February 23, 2026, https://web.gps.caltech.edu/users/jess/publications_files/ChenGCA2018.pdf
7. (PDF) Ion partitioning and the geochemistry of coral skeletons: Solving the mystery of the vital effect - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/260292462_Ion_partitioning_and_the_geochemistry_of_coral_skeletons_Solving_the_mystery_of_the_vital_effect
8. Stable isotopes ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) of organic matrix from coral skeleton - PMC - NIH, accessed February 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC547863/>
9. Coral carbon isotope sensitivity to growth rate and water depth with ..., accessed February 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6499886/>
10. Vital effects in coral skeletal composition display strict three- dimensional control, accessed February 23, 2026, https://www.researchgate.net/publication/232731052_Vital_effects_in_coral_skeletal_composition_display_strict_three-dimensional_control
11. Variability of skeletal growth and $\delta^{13}\text{C}$ in massive corals from the South China Sea: Effects of photosynthesis, respiration and human activities | Request PDF - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/229420660_Variability_of_skeletal_growth_and_d13C_in_massive_corals_from_the_South_China_Sea_Effects_of_photosynthesis_respiration_and_human_activities
12. Ocean acidification in the Western Pacific: Boron isotopic composition recorded in a tropical massive coral core from Lanyu Islet SE Taiwan - Frontiers, accessed February 23, 2026, <https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2022.877810/full>
13. Kinetic and metabolic isotope effects in coral skeletal carbon isotopes: A re-evaluation using experimental coral bleaching as a case study - the UWA Profiles and Research Repository, accessed February 23, 2026, <https://research-repository.uwa.edu.au/en/publications/kinetic-and-metabolic-isotope-effects-in-coral-skeletal-carbon-is/>
14. Twentieth century sea surface temperature and salinity ... - OceanRep, accessed February 23, 2026, <https://oceanrep.geomar.de/id/eprint/25697/1/Cahyarini%20et.al..pdf>
15. The CoralHydro2k database: a global, actively curated compilation of coral $\delta^{18}\text{O}$ and Sr/Ca proxy - ESSD Copernicus, accessed February 23, 2026,

- <https://essd.copernicus.org/articles/15/2081/2023/essd-15-2081-2023.pdf>
16. Sr/Ca and $\delta^{18}\text{O}$ in a fast-growing *Diploria strigosa* coral: Evaluation of a new climate archive for the tropical Atlantic - OceanRep - geomar, accessed February 23, 2026, https://oceanrep.geomar.de/257/1/2006_Hetzing_2006GC001347.pdf
 17. Sea Surface Temperature and Salinity in Lombok Strait Reconstructed From Coral Sr/Ca and $\delta^{18}\text{O}$, 1962–2012 - Frontiers, accessed February 23, 2026, <https://www.frontiersin.org/journals/climate/articles/10.3389/fclim.2022.918273/full>
 18. The PAGES CoralHydro2k Seawater $\delta^{18}\text{O}$ Database - ESSD Copernicus, accessed February 23, 2026, <https://essd.copernicus.org/preprints/essd-2025-467/essd-2025-467.pdf>
 19. Machine Learning Solutions to Regional Surface Ocean $\delta^{18}\text{O}$ -Salinity Relationships for Paleoclimatic Reconstruction - Archimer, accessed February 23, 2026, <https://archimer.ifremer.fr/doc/00884/99628/109651.pdf>
 20. Environmental Records from Great Barrier Reef Corals: Inshore ..., accessed February 23, 2026, <https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0077091&type=printable>
 21. FOUNDATION ELEMENTS - Red Sea Reef Care Program, accessed February 23, 2026, https://g1.redseafish.com/reef-care-program/reef_foundation_program/
 22. Trace element proxies and stable isotopes used to identify water quality threats to elkhorn coral (*Acropora palmata*) at two nati - NOAA Central Library, accessed February 23, 2026, https://library.oarcloud.noaa.gov/noaa_documents.lib/IR_articles/FY2023_Submissions/Manuscripts/BaylessA_MPB_2021_manu.pdf
 23. Sea surface temperature and salinity reconstruction from coral geochemical tracers, accessed February 23, 2026, <https://repository.geologyscience.ru/server/api/core/bitstreams/e5d95b94-768c-45f6-bdc1-a1eaf388c4b1/content>
 24. Paired coral Sr/Ca and $\delta^{18}\text{O}$ records reveal increasing ENSO influence on Malaysian Borneo's hydroclimate - OceanRep, accessed February 23, 2026, <https://oceanrep.geomar.de/60835/1/1-s2.0-S0048969724071006-main.pdf>
 25. Sea surface temperature and salinity reconstruction from coral geochemical tracers | Request PDF - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/222394327_Sea_surface_temperature_and_salinity_reconstruction_from_coral_geochemical_tracers
 26. Regime shift of skeletal $\delta^{13}\text{C}$ after 1997/1998 El Niño event in *Porites* coral from Green Island, Taiwan - PMC, accessed February 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11458579/>
 27. Coral Ba/Ca molar ratios as a proxy of precipitation in the northern Yucatan Peninsula, Mexico | Request PDF - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/257266001_Coral_BaCa_molar_ratios_as_a_proxy_of_precipitation_in_the_northern_Yucatan_Peninsula_Mexico
 28. Multi-colony calibrations of coral Ba/Ca with a contemporaneous in situ seawater barium record, accessed February 23, 2026, <https://par.nsf.gov/servlets/purl/10214818>

29. Coral skeletal proxy records database for the Great Barrier Reef, Australia, accessed February 23, 2026, <https://essd.copernicus.org/articles/16/4869/2024/>
30. Deconvolving the biogeochemical controls on coral Sr ... - EGU sphere, accessed February 23, 2026, <https://egusphere.copernicus.org/preprints/2025/egusphere-2025-4126/egusphere-2025-4126.pdf>
31. Mg/Ca Thermometry in Coral Skeletons - PubMed, accessed February 23, 2026, <https://pubmed.ncbi.nlm.nih.gov/8875933/>
32. Empirical assessment of coral Sr/Ca and Mg/Ca ratios as climate proxies using colonies grown at different temperatures | Request PDF - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/241062761_Empirical_assessment_of_coral_SrCa_and_MgCa_ratios_as_climate_proxies_using_colonies_grown_at_different_temperatures
33. Mg structural state in coral aragonite and implications for the paleoenvironmental proxy, accessed February 23, 2026, https://www.researchgate.net/publication/251429038_Mg_structural_state_in_coral_aragonite_and_implications_for_the_paleoenvironmental_proxy
34. Separation of kinetic and metabolic isotope effects in carbon-13 records preserved in reef coral skeletons | Request PDF - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/publication/221988373_Separation_of_kinetic_and_metabolic_isotope_effects_in_carbon-13_records_preserved_in_reef_coral_skeletons
35. A model of calcification (following Adkins et al. 2003). CO₂ and water... - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/figure/A-model-of-calcification-following-Adkins-et-al-2003-CO-2-and-water-diffuse-into-the_fig4_254823130
36. Short-Term Coral Bleaching Is Not Recorded by Skeletal Boron Isotopes - PMC, accessed February 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4232377/>
37. Coral calcifying fluid pH dictates response to ocean acidification, accessed February 23, 2026, <https://www.vliz.be/imisdocs/publications/258113.pdf>
38. A re-appraisal of the ENSO response to volcanism with paleoclimate data assimilation - the NOAA Institutional Repository, accessed February 23, 2026, https://repository.library.noaa.gov/view/noaa/62366/noaa_62366_DS1.pdf
39. Corals Make Reliable Recorders of El Niño Fluctuations - Eos.org, accessed February 23, 2026, <https://eos.org/research-spotlights/corals-make-reliable-recorders-of-el-nino-fluctuations>
40. Absolute correlations between coral Sr/Ca and $\delta^{18}\text{O}$ and local sea... - ResearchGate, accessed February 23, 2026, https://www.researchgate.net/figure/Absolute-correlations-between-coral-Sr-Ca-and-d18O-and-local-sea-surface-temperature_fig4_371325750
41. Paleo Data Search | Study | National Centers for Environmental Information (NCEI) - NOAA, accessed February 23, 2026,

- <https://www.ncei.noaa.gov/access/paleo-search/study/25891>
42. On the variability of ENSO over the past six centuries - Penn State, accessed February 23, 2026,
<https://pure.psu.edu/en/publications/on-the-variability-of-enso-over-the-past-six-centuries/>
 43. Global Multiproxy ENSO Reconstruction Over the Past Millennium - ResearchGate, accessed February 23, 2026,
https://www.researchgate.net/publication/380638582_Global_Multiproxy_ENSO_Reconstruction_Over_the_Past_Millennium
 44. Dual clumped isotope thermometry of coral carbonate | Request PDF - ResearchGate, accessed February 23, 2026,
https://www.researchgate.net/publication/364532999_Dual_clumped_isotope_the_thermometry_of_coral_carbonate
 45. Clumped Isotope Paleothermometry: Principles, Applications, and Challenges | The Paleontological Society Papers | Cambridge Core, accessed February 23, 2026,
<https://www.cambridge.org/core/journals/the-paleontological-society-papers/article/clumped-isotope-paleothermometry-principles-applications-and-challenges/3B8E1B957C6EFC8CA167902623CDA7C7>
 46. UH Supercomputer Study Shows Coral Can't Flee Warming, But Urgent Action Can Still Protect Them, accessed February 23, 2026,
<https://www.himb.hawaii.edu/2025/06/06/uh-supercomputer-study-shows-coral-cant-flee-warming-but-urgent-action-can-still-protect-them/>
 47. Coral collapse signals Earth's first climate tipping point - Science News, accessed February 23, 2026,
<https://www.sciencenews.org/article/coral-collapse-climate-tipping-point>
 48. SF scientists fight climate doom with lab-grown coral breakthrough - SFGATE, accessed February 23, 2026,
<https://www.sfgate.com/bayarea/article/sf-scientists-fight-climate-doom-breakthrough-21041729.php>
 49. Projections of coral reef carbonate production from a global climate–coral reef coupled model - BG, accessed February 23, 2026,
<https://bg.copernicus.org/articles/22/4531/2025/>
 50. Experimental coral reef communities transform yet persist under mitigated future ocean warming and acidification | PNAS, accessed February 23, 2026,
<https://www.pnas.org/doi/10.1073/pnas.2407112121>